

PinIT Career OS — 36+ Individual 1-Month Courses Master Visuals Architecture

Document Identifier	PINIT_36_INDIVIDUAL_COURSES_VISUALS_PLAN (Claude Style Publication Edition)
Total Courses & Volume	37 Courses • 1,110 Lesson Days • 6,660 Interactive Lesson Parts
Curriculum Domain Clusters	6 Distinct Clusters (Software Eng, Systems/Security, Hardware/Web3, Commerce, Operations, Foundations)
Quality & Testing Standard	Technical Gate v2 (Rules R1–R14) with 5-Day Atomic Verification Cycles & Zero Hallucination
Engine & API Stack	Next.js 14, TypeScript, React VisualStage.tsx , /api/visuals , Playwright Headless Chromium

1. Executive Summary & The Golden Triad

The PinIT Career OS learning platform delivers high-retention technical and vocational education by pairing every lesson part with an interactive visual mental model. Across all **37 individual 1-month certification courses**, abstract code, business logic, hardware registers, and algorithmic structures are materialized on the left of the student's lesson player.

The Golden Triad: Content ↔ Output ↔ Visuals. Visuals move in lockstep with the instructor's spoken voice (say lines). Visual values are **strictly bound to runtime compiler output**, never hallucinated by an AI. When students edit code and click Run, diagrams smoothly re-animate using their live values.

2. Master Domain Cluster Taxonomy (37 Courses)

CLUSTER-1: SE-WEB-CLOUD

Core Software Engineering, Web & Cloud Systems

#	Course Identifier	Canonical Title	Level	Allowed Templates
01	course-java-logic	Java Fundamentals & Core Logic	Beginner	boxes, flow, table, cells, stack-queue, compare
02	course-react-web	Full-Stack React Web Development	Intermediate	component-tree, wireframe, flow, boxes, compare, states
03	course-node-web	Node.js & TypeScript Backend Engineering	Intermediate	sequence, stack-queue, flow, boxes, table, states
04	course-cloud-native	Cloud Native Architectures (AWS)	Advanced	workflow, sequence, states, table, bars, compare
05	course-devops-cicd	DevOps & CI/CD Pipeline Automation	Advanced	flow, workflow, states, sequence, compare

#	Course Identifier	Canonical Title	Level	Allowed Templates
06	course-design-systems	UI/UX Design Systems & Visual Frontend	Beginner	wireframe, component-tree, compare, boxes, bars

CLUSTER-2: SYS-SEC-DATA

Systems, Mobile, Security, DB & Distributed Systems

#	Course Identifier	Canonical Title	Level	Allowed Templates
07	course-dsa-optim	Data Structures & Algorithmic Optimizations	Intermediate	cells, stack-queue, tree-graph, table, bars, compare
08	course-mobile-dev	Mobile Application Development	Intermediate	wireframe, component-tree, flow, sequence, states
09	course-cybersecurity	Cybersecurity Principles & Secure Systems	Advanced	flow, sequence, table, compare, states
10	course-database-eng	Database Engineering & Query Performance	Advanced	table, tree-graph, flow, compare, bars
11	course-distributed-sys	High-Scale Distributed System Design	Advanced	sequence, workflow, states, table, cells, compare
12	course-ai-eng	AI Engineering & LLM Integration	Intermediate	flow, table, bars, sequence, compare, boxes
13	course-fullstack-js	Full-Stack JavaScript Engineering	Intermediate	flow, component- tree, sequence, table, states, compare

CLUSTER-3: EMBEDDED-QUANT

Embedded Hardware, 3D, Web3 & High-Frequency Systems

#	Course Identifier	Canonical Title	Level	Allowed Templates
14	course-iot-embedded	IoT, Firmware & Embedded Systems	Intermediate	flow, states, cells, boxes, sequence, register- bits
15	course-3d-graphics	3D Interactive Graphics & Avatar Animation	Advanced	tree-graph, boxes, flow, compare, bars
16	course-blockchain-web3	Blockchain, Web3 & Smart Contracts	Intermediate	tree-graph, stack- queue, flow, sequence, compare, states

#	Course Identifier	Canonical Title	Level	Allowed Templates
17	course-iot-network	IoT Wireless Networks & Protocols	Intermediate	flow, sequence, states, table, compare
18	course-iot-edge-ai	Edge AI, DSP & TinyML Systems	Advanced	flow, bars, cells, table, compare
19	course-iot-security	Industrial IoT Security & Device Lifecycle	Advanced	flow, sequence, states, compare, table
20	course-python-backend	Python Programming & Backend Systems	Intermediate	flow, sequence, boxes, table, states, compare
21	course-quant-systems	Quantitative Engineering & Low-Latency Trading Systems	Advanced	stack-queue, table, bars, flow, cells, compare

CLUSTER-4: FIN-ACCT-BIZ**Digital Accounting, Corporate Finance & Business Analytics**

#	Course Identifier	Canonical Title	Level	Allowed Templates
22	course-digital-accounting	Digital Accounting & Taxation (B.Com / BBA)	Beginner	table, flow, compare, bars, ledger-sheet
23	course-finance-investment	Business Finance & Investment Management (B.Com / BBA)	Beginner	table, bars, flow, compare
24	course-business-analytics	Business Analytics & Decision Intelligence (B.Com / BBA / MBA)	Beginner	bars, table, flow, compare
25	course-marketing-branding	Marketing & Brand Management (B.Com / BBA / MBA)	Beginner	flow, compare, table, bars
26	course-digital-marketing	Digital Marketing & Growth Strategy (B.Com / BBA / MBA)	Beginner	flow, bars, table, compare, funnel

CLUSTER-5: ECOM-OPS-AI**Digital Commerce, Operations, Sales & AI Transformation**

#	Course Identifier	Canonical Title	Level	Allowed Templates
27	course-ecommerce-digital-biz	E-Commerce & Digital Business (B.Com / BBA / MBA)	Beginner	flow, table, sequence, bars, compare, funnel
28	course-entrepreneurship-biz-mgmt	Entrepreneurship & Business Management (B.Com / BBA / MBA)	Beginner	flow, table, compare, bars
29	course-sales-crm-success	Sales, Customer Success & CRM (B.Com / BBA / MBA)	Beginner	flow, table, states, bars, compare, funnel
30	course-operations-supplychain-compliance	Operations, Supply Chain & Business Compliance (B.Com / BBA / MBA)	Beginner	flow, table, bars, compare

#	Course Identifier	Canonical Title	Level	Allowed Templates
31	course-ai-digital-transformation	AI & Digital Transformation for Business (B.Com / BBA / MBA)	Beginner	workflow, flow, table, compare, bars

CLUSTER-6: FOUNDATIONS

Universal Digital Foundations, Tools & Cognitive Communication

#	Course Identifier	Canonical Title	Level	Allowed Templates
32	course-computer-fundamentals	Computer Literacy, Digital Productivity & OS Fundamentals	Beginner	flow, boxes, tree-graph, table, compare
33	course-ai-prompt-literacy	Everyday AI Literacy & Prompt Engineering	Beginner	flow, compare, boxes, table, bars
34	course-excel-data-viz	Excel & Data Analysis Fundamentals	Beginner	table, cells, bars, compare, flow
35	course-git-version-control	Git, GitHub & Version Control Basics	Beginner	tree-graph, flow, states, compare, boxes
36	course-softskills-communication	Professional Tech Communication & Interview Mastery	Beginner	flow, compare, table, states
37	course-nlp	Natural Language Processing & Computational Linguistics	Advanced	letters, cells, table, flow, bars, tree-graph

3. Master Visual Templates Catalog (17 Interactive Diagram Engines)

The visual platform provides 17 dedicated diagram engines. Each engine complies with the 4 core design tokens (`data` , `ok` , `error` , `idle`) and supports animated step transitions:

Template Key	Engine Name	Domain	Interactive Mechanics & Pedagogical Function
<code>flow</code>	Multi-Stage Flow	Universal	Linear and branching pipelines with stage status highlights
<code>boxes</code>	Memory & State Boxes	Systems / Web	Slots holding named variables, props, register values, or configs
<code>table</code>	Relational & Ledger Table	Data / Fin / API	Columns and rows with row-by-row or cell-by-cell step illumination
<code>letters</code>	Character Array	CompSci / NLP	Character cells with moving pointers for strings, tokens, and regex
<code>compare</code>	Comparative Panels	Universal	Side-by-side comparison with pass/fail indicators and pros/cons
<code>cells</code>	1D Memory Array Buffer	DSA / Hardware	Indexed array blocks with up to 3 named moving pointers
<code>stack-queue</code>	LIFO Stack / FIFO Queue	Systems / Trading	Vertical stack frames or horizontal queue with push/pop animations
<code>tree-graph</code>	Hierarchical Tree / Graph	DSA / Web / 3D	Up to 6 nodes with directional edges and traversal highlighting
<code>bars</code>	Quantitative Metrics Bars	Analytics / SRE	Labeled numeric bars comparing latency, ratios, or benchmarks
<code>sequence</code>	Chronological Sequence	Network / Auth	Lifeline columns with directional request/response message arrows
<code>states</code>	Finite State Machine	Protocols / Async	State bubbles with transitions and active state illumination
<code>workflow</code>	Multi-Tier Topology	Cloud / DevOps	System architecture topology connecting client, proxy, pods, DB
<code>component-tree</code>	UI Component Hierarchy	Frontend / Mobile	Component tree displaying props cascading down and events bubbling up
<code>wireframe</code>	Box-Model Wireframe	Frontend / Design	Nested UI boxes visualizing layout engines, margin, border, padding
<code>register-bits</code>	Bitwise Register Simulator	Embedded / IoT	8/16/32-bit register cells visualizing bitmasks, set/clear operations
<code>ledger-sheet</code>	Double-Entry T-Account	Commerce / Acct	T-Account debit/credit columns maintaining balancing invariants
<code>funnel</code>	Conversion Funnel	Marketing / Sales	Multi-stage conversion stages showing drop-off percentages and cohorts

4. Technical Gate v2 Enforcement Standard (Rules R1–R14)

To guarantee **zero AI hallucinations** and production resilience, every generated JSON visual file must pass all 14 rules of the Gate v2 programmatic validator:

Rule	Rule Name	Validation Criterion
R1	AST Linkage	Day file has exactly 6 entries matching real lesson AST part titles.
R2	Template Whitelist	Template is <code>none</code> or on the course's allowed list.
R3	Shape & Step Limits	2 to 5 steps; ≤ 6 shapes per step; tables have 2–5 cols and ≤ 6 rows.
R4	Monotonic Stepping	Step <code>at</code> points strictly increase and match real lecture sentence indices.
R5	Caption Typography	Single sentence ≤ 80 characters, ending in a period, zero emojis.
R6	Deterministic Bindings	Every displayed value maps to a compiler runtime binding (<code>var</code> , <code>out</code> , etc.).
R7	Numeric Grounding	Every number in a caption appears verbatim in code or runtime output.
R8	Lexical Grounding	Every tapped label exists as a whole word in lesson text or code.
R9	Design Token Palette	Tones restricted strictly to <code>data</code> , <code>ok</code> , <code>error</code> , <code>idle</code> .
R10	Visual Density	At least 3 of 6 parts per day have diagrams (otherwise flagged review).
R11	Manifest Sync	Manifest status reflects true file state (<code>passed</code> , <code>none</code> , <code>needs-review</code>).
R12	Concept Overlap	Captions share 4+ letter keywords with attached lesson explanation.
R13	Code Freshness	<code>codeHash</code> matches SHA-256 hash of current lesson snippet.
R14	Runtime Stability	Bindings point only to stable variables that never fluctuate across runs.

5. Exhaustive Course-by-Course Blueprint (All 37 Courses)

Below are the complete architectural blueprints for all 37 individual 1-month courses. Each course specifies its domain cluster, allowed templates, pedagogical mental model, 6-block curriculum breakdown (Days 1–30), and capstone project:

Course 01: Java Fundamentals & Core Logic

Catalog: `course-java-logic`

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: `boxes`, `flow`, `table`, `cells`, `stack-queue`, `compare`

Architectural Theme: JVM Architecture, Primitive Typing, OOP Invariants & Enterprise Modularity

***Pedagogical Whiteboard Model:** Illustrate JVM Stack frame pushes/pops, heap reference allocations, and variable mutations during loops. Visuals clarify value vs reference semantics.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	JVM Bytecode execution, Primitive boxing & System.out printing flow	Template <code>boxes</code> : Animated state progression
Block 2	Days 6–10	Conditional branching, Relational operators & Boolean logic tables	Template <code>flow</code> : Animated state progression
Block 3	Days 11–15	While/Do-While loops, Iterative counters & Array indexing cells	Template <code>table</code> : Animated state progression
Block 4	Days 16–20	For loops, Nested iteration patterns & Matrix coordinate grids	Template <code>cells</code> : Animated state progression
Block 5	Days 21–25	Method call frames, Parameter pass-by-value & Return value routing	Template <code>stack-queue</code> : Animated state progression
Block 6	Days 26–30	OOP Class blueprints, Encapsulation & Milestone Financial Engine	Template <code>compare</code> : Animated state progression

Capstone Milestone: Milestone Financial Utility Engine: A multi-module Java application modeling bank accounts, compound interest calculators, and transactional debit/credit validation.

Course 02: Full-Stack React Web Development

Catalog: `course-react-web`

Difficulty: Intermediate | **Duration:** 4 Weeks (30 Days / 6 Blocks) | **Quests:** 96 quests | **Allowed:** `component-tree`, `wireframe`, `flow`, `boxes`, `compare`, `states`

Architectural Theme: Fiber Reconciliation, Hook Closures, Virtual DOM Diffing & Component Trees

Pedagogical Whiteboard Model: Show component trees with props cascading downwards and callback events bubbling up. Highlight component re-renders triggered by state setter dispatches.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Declarative JSX parsing, Component composition & Virtual DOM tree	Template <code>component-tree</code> : Animated state progression
Block 2	Days 6–10	useState hook closures, Immutable state updates & Re-render triggers	Template <code>wireframe</code> : Animated state progression
Block 3	Days 11–15	Props contract interfaces, Prop drilling vs Context & Children nodes	Template <code>flow</code> : Animated state progression
Block 4	Days 16–20	useEffect dependency arrays, Async fetch cycles & Cleanup timers	Template <code>boxes</code> : Animated state progression
Block 5	Days 21–25	Controlled forms, Validation state machines & UI layout wireframes	Template <code>compare</code> : Animated state progression
Block 6	Days 26–30	Next.js App Router hybrid rendering, Server vs Client components	Template <code>states</code> : Animated state progression

Capstone Milestone: Production Enterprise SaaS Portal: High-performance Next.js application with dynamic dashboard routing, dark/light theme tokens, and live analytics telemetry.

Course 03: Node.js & TypeScript Backend Engineering

Catalog: course-node-web

Difficulty: Intermediate | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: sequence, stack-queue, flow, boxes, table, states

Architectural Theme: Asynchronous Event Loop, Middleware Pipelines & Relational Microservices

Pedagogical Whiteboard Model: Map incoming HTTP request flows through Express middleware waterfalls. Show database connection pool acquisitions and async non-blocking I/O queues.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	V8 Engine Call Stack, libuv Event Loop & Microtask/Macrotask queues	Template sequence : Animated state progression
Block 2	Days 6–10	TypeScript strict typing, Interface schemas & Compile-time validation	Template stack-queue : Animated state progression
Block 3	Days 11–15	HTTP Request-Response lifecycles, RESTful routes & Express middleware	Template flow : Animated state progression
Block 4	Days 16–20	Relational PostgreSQL queries, Connection pooling & ACID transactions	Template boxes : Animated state progression
Block 5	Days 21–25	JWT HMAC-SHA256 authentication, Bcrypt password hashing & Auth guards	Template table : Animated state progression
Block 6	Days 26–30	Containerized production Docker build & Health-check endpoint telemetry	Template states : Animated state progression

Capstone Milestone: High-Concurrency REST Microservice: Authenticated backend API with role-based access control, connection-pooled PostgreSQL persistence, and Docker multi-stage deployment.

Course 04: Cloud Native Architectures (AWS)

Catalog: course-cloud-native

Difficulty: Advanced | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: workflow, sequence, states, table, bars, compare

Architectural Theme: AWS VPC Topologies, Serverless Lifecycles & Highly Available Cloud Systems

Pedagogical Whiteboard Model: Visualize cloud architecture topologies: VPC subnets, Internet Gateways, Application Load Balancers, and ECS tasks. Clarify IAM role evaluation logic.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	AWS Global Cloud Infrastructure, Regions, Availability Zones & IAM Policies	Template workflow : Animated state progression
Block 2	Days 6–10	VPC CIDR subnets, Public vs Private routing & Security Group firewalls	Template sequence : Animated state progression
Block 3	Days 11–15	Compute scaling: EC2 instances, Application Load Balancers & Auto Scaling	Template states : Animated state progression
Block 4	Days 16–20	Serverless computing: AWS Lambda triggers, API Gateway & DynamoDB NoSQL	Template table : Animated state progression
Block 5	Days 21–25	Container orchestration: ECR image registries & ECS/Fargate task definitions	Template bars : Animated state progression
Block 6	Days 26–30	Multi-AZ High Availability, Route53 DNS failover & Terraform IaC blueprints	Template compare : Animated state progression

Capstone Milestone: Multi-AZ Serverless Cloud Infrastructure: Terraform-provisioned VPC architecture with ALB traffic distribution, ECS Fargate microservices, and automated health checks.

Course 05: DevOps & CI/CD Pipeline Automation

Catalog: `course-devops-cicd`

Difficulty: Advanced | **Duration:** 4 Weeks (30 Days / 6 Blocks) | **Quests:** 96 quests | **Allowed:** `flow`, `workflow`, `states`, `sequence`, `compare`

Architectural Theme: Immutable Containerization, Automated CI/CD Pipelines & Kubernetes Orchestration

Pedagogical Whiteboard Model: *Diagram Docker layer caching trees, GitHub Actions stage execution waterfalls, and Kubernetes ReplicaSet pod reconciliation loops.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	POSIX Linux administration, Shell piping & Systemd daemon services	Template <code>flow</code> : Animated state progression
Block 2	Days 6–10	Git trunk-based branching, Semantic versioning & GitOps workflows	Template <code>workflow</code> : Animated state progression
Block 3	Days 11–15	Docker engine internals, Multi-stage image builds & Layer caching DAGs	Template <code>states</code> : Animated state progression
Block 4	Days 16–20	GitHub Actions runners, Lint-Test-Build matrix workflows & Secret vaults	Template <code>sequence</code> : Animated state progression
Block 5	Days 21–25	Kubernetes cluster architecture: Control plane, Pods, Services & Ingress	Template <code>compare</code> : Animated state progression
Block 6	Days 26–30	Zero-downtime Blue/Green deployments, Canary rollouts & Prometheus alerts	Template <code>flow</code> : Animated state progression

Capstone Milestone: Enterprise Automated GitOps Pipeline: Multi-stage Docker containerization pipeline with automated GitHub Actions testing, security vulnerability scanning, and Kubernetes deployment.

Course 06: UI/UX Design Systems & Visual Frontend

Catalog: course-design-systems

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: wireframe, component-tree, compare, boxes, bars

Architectural Theme: Atomic Component Architectures, Design Tokens & Accessible Web Layouts

Pedagogical Whiteboard Model: Draw CSS box-model margins/borders/padding, Flexbox/Grid spatial coordinates, and accessibility contrast verification ratios across light and dark modes.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Visual hierarchy principles, Modular typography scales & 8pt grid geometry	Template wireframe : Animated state progression
Block 2	Days 6–10	Color space tokens, HSL palette generation & WCAG AAA contrast ratios	Template component-tree : Animated state progression
Block 3	Days 11–15	Atomic Design methodology: Atoms, Molecules, Organisms & Templates	Template compare : Animated state progression
Block 4	Days 16–20	CSS Layout mechanics: Flexbox axes, CSS Grid fr tracks & Responsive breakpoints	Template boxes : Animated state progression
Block 5	Days 21–25	Interactive component state machines: Hover, Focus-visible, Active & Disabled	Template bars : Animated state progression
Block 6	Days 26–30	Design Token architecture: Style Dictionary, JSON token exports & Figma sync	Template wireframe : Animated state progression

Capstone Milestone: Production Design System Library: Documented, accessible UI component kit featuring tokens, atomic components, and responsive mobile-to-desktop wireframe layouts.

Course 07: Data Structures & Algorithmic Optimizations

Catalog: course-dsa-optim

Difficulty: Intermediate | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: cells, stack-queue, tree-graph, table, bars, compare

Architectural Theme: Asymptotic Complexity, Memory Buffers, Tree Traversal & Dynamic Programming

Pedagogical Whiteboard Model: Show memory array buffers with dual pointers shifting, recursion stack frame winding/unwinding, and 2D memoization grids populating optimal substructure solutions.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Big-O Asymptotic Space-Time analysis, Linear arrays & In-place operations	Template cells : Animated state progression
Block 2	Days 6–10	Two-pointer convergence, Sliding window bounds & Prefix sum caching	Template stack-queue : Animated state progression
Block 3	Days 11–15	Linked memory lists, LIFO Call Stacks & FIFO Circular Queue buffers	Template tree-graph : Animated state progression
Block 4	Days 16–20	Hierarchical Trees, Binary Search Tree balancing & Binary Heap priorities	Template table : Animated state progression
Block 5	Days 21–25	Graph topologies, Adjacency lists, BFS/DFS traversal & Dijkstra shortest paths	Template bars : Animated state progression
Block 6	Days 26–30	Dynamic Programming: Memoization tables, Tabulation & Knapsack optimization	Template compare : Animated state progression

Capstone Milestone: High-Performance Algorithmic Optimization Suite: Benchmark-validated implementations of cache-conscious searching, graph pathfinding, and dynamic programming schedulers.

Course 08: Mobile Application Development

Catalog: `course-mobile-dev`

Difficulty: Intermediate | **Duration:** 4 Weeks (30 Days / 6 Blocks) | **Quests:** 96 quests | **Allowed:** `wireframe`, `component-tree`, `flow`, `sequence`, `states`

Architectural Theme: Cross-Platform React Native, JSI Runtime, Native Modules & Offline SQLite

Pedagogical Whiteboard Model: *Illustrate JavaScript-to-Native bridge execution, Yoga Flexbox layout reflows, and touch gesture responder state transitions.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	React Native architecture: JavaScript thread, JSI, Shadow Tree & Native UI	Template <code>wireframe</code> : Animated state progression
Block 2	Days 6–10	Core UI primitives: View, Text, FlatList virtualization & Yoga flex layouts	Template <code>component-tree</code> : Animated state progression
Block 3	Days 11–15	Navigation architectures: Native Stack, Bottom Tabs & Deep-link URI routing	Template <code>flow</code> : Animated state progression
Block 4	Days 16–20	Offline state persistence: MMKV fast key-value & SQLite local transactions	Template <code>sequence</code> : Animated state progression
Block 5	Days 21–25	Native hardware bridges: Camera capture, GPS Geolocation & Biometric auth	Template <code>states</code> : Animated state progression
Block 6	Days 26–30	Performance profiling: Hermes bytecode compilation & Release bundle packaging	Template <code>wireframe</code> : Animated state progression

Capstone Milestone: Cross-Platform Mobile Utility App: Feature-complete React Native application featuring virtualized infinite feeds, offline SQLite synchronization, and biometric authentication.

Course 09: Cybersecurity Principles & Secure Systems

Catalog: course-cybersecurity

Difficulty: Advanced | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: flow, sequence, table, compare, states

Architectural Theme: Cryptographic Protocols, OWASP Top 10 Mitigation & Zero-Trust Defense

Pedagogical Whiteboard Model: Diagram packet inspection sequences, TLS 1.3 cryptographic key exchanges, JWT tampering detection, and SQL injection sanitization barriers.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Threat modeling methodologies, Attack vectors & OWASP Top 10 vulnerabilities	Template flow : Animated state progression
Block 2	Days 6–10	Authentication architecture: Multi-factor auth, Salted Bcrypt & Session stores	Template sequence : Animated state progression
Block 3	Days 11–15	Cryptographic systems: Symmetric AES-GCM, Asymmetric RSA & ECC key pairs	Template table : Animated state progression
Block 4	Days 16–20	Web application hardening: CSP headers, CORS policies & CSRF token validation	Template compare : Animated state progression
Block 5	Days 21–25	Network defense: TLS 1.3 handshake, WAF firewall rules & DDoS rate limiters	Template states : Animated state progression
Block 6	Days 26–30	Security incident response: Penetration testing workflows & SIEM audit logs	Template flow : Animated state progression

Capstone Milestone: Zero-Trust Secure Authentication Gateway: End-to-end hardened authentication microservice with Argon2id password hashing, rotating JWT signing keys, and brute-force rate limiting.

Course 10: Database Engineering & Query Performance

Catalog: course-database-eng

Difficulty: Advanced | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: table, tree-graph, flow, compare, bars

Architectural Theme: Relational Theory, B+ Tree Indexes, Transaction Isolation & WAL Logging

Pedagogical Whiteboard Model: Illustrate B+ Tree node splits, Write-Ahead Log (WAL) sequential flushes, and query execution plans comparing sequential table scans with index lookups.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Relational algebra, Entity-Relationship schemas & Normalization (1NF–BCNF)	Template table : Animated state progression
Block 2	Days 6–10	SQL Query execution engine: Parsing, Query planning & AST optimization	Template tree-graph : Animated state progression
Block 3	Days 11–15	Indexing data structures: B+ Tree page balancing, Hash indexes & Covering indexes	Template flow : Animated state progression
Block 4	Days 16–20	ACID properties, MVCC concurrency control & Transaction isolation anomalies	Template compare : Animated state progression
Block 5	Days 21–25	Query profiling: EXPLAIN ANALYZE interpretation & Buffer cache hit ratios	Template bars : Animated state progression
Block 6	Days 26–30	Horizontal scaling: Table partitioning, Read replication & Connection pools	Template table : Animated state progression

Capstone Milestone: High-Throughput Financial Ledger Database: Production-tuned PostgreSQL schema with B+ tree composite indexes, strict MVCC isolation, and sub-millisecond query execution plans.

Course 11: High-Scale Distributed System Design

Catalog: `course-distributed-sys`

Difficulty: Advanced | **Duration:** 4 Weeks (30 Days / 6 Blocks) | **Quests:** 96 quests | **Allowed:** `sequence`, `workflow`, `states`, `table`, `cells`, `compare`

Architectural Theme: Consensus Protocols, Consistent Hashing, Distributed State & Fault Tolerance

***Pedagogical Whiteboard Model:** Show consistent hashing token rings, Raft leader election heartbeat sequences, and 2-Phase Commit prepare/commit transaction message flows.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Fallacies of distributed computing, Network partitions & CAP/PACELC theorems	Template <code>sequence</code> : Animated state progression
Block 2	Days 6–10	RPC protocols, Protocol Buffers schema compilation & gRPC streaming channels	Template <code>workflow</code> : Animated state progression
Block 3	Days 11–15	Data partitioning: Consistent hashing rings, Virtual nodes & Key distribution	Template <code>states</code> : Animated state progression
Block 4	Days 16–20	Consensus algorithms: Paxos & Raft leader election, Term leases & Log commits	Template <code>table</code> : Animated state progression
Block 5	Days 21–25	Distributed transactions: Two-Phase Commit (2PC) & Saga compensating patterns	Template <code>cells</code> : Animated state progression
Block 6	Days 26–30	Resilience patterns: Circuit breakers, Distributed rate limiting & Jaeger tracing	Template <code>compare</code> : Animated state progression

Capstone Milestone: Fault-Tolerant Distributed Key-Value Store: Raft-replicated cluster featuring automated leader failover, consistent hashing shard allocation, and vector clock conflict detection.

Course 12: AI Engineering & LLM Integration

Catalog: course-ai-eng

Difficulty: Intermediate | **Duration:** 4 Weeks (30 Days / 6 Blocks) | **Quests:** 96 quests | **Allowed:** flow, table, bars, sequence, compare, boxes

Architectural Theme: Large Language Model Systems, Vector Similarity Search, RAG & Autonomous Agents

Pedagogical Whiteboard Model: Draw token embedding coordinate spaces, RAG vector retrieval cosine similarity rankings, and agentic tool dispatching request/response loops.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Transformer architecture fundamentals: Tokenization, Embeddings & Self-Attention	Template flow : Animated state progression
Block 2	Days 6–10	Prompt engineering patterns: Few-shot context, System framing & Chain-of-Thought	Template table : Animated state progression
Block 3	Days 11–15	Vector databases: Dense embeddings, HNSW indexing & Cosine similarity search	Template bars : Animated state progression
Block 4	Days 16–20	Retrieval-Augmented Generation (RAG): Document chunking & Context injection	Template sequence : Animated state progression
Block 5	Days 21–25	Agentic tool calling: JSON function signatures, API dispatch & ReAct loops	Template compare : Animated state progression
Block 6	Days 26–30	Evaluation & Guardrails: Hallucination detection, Token budgets & Model routing	Template boxes : Animated state progression

Capstone Milestone: Enterprise Document Intelligence Copilot: Production RAG pipeline with semantic document chunking, hybrid vector search, citation synthesis, and guardrail verification.

Course 13: Full-Stack JavaScript Engineering

Catalog: `course-fullstack-js`

Difficulty: Intermediate | **Duration:** 4 Weeks (30 Days / 6 Blocks) | **Quests:** 384 quests | **Allowed:** `flow`, `component-tree`, `sequence`, `table`, `states`, `compare`

Architectural Theme: Monolithic to Microservice JS, Next.js Server Components, ORMs & WebSockets

Pedagogical Whiteboard Model: Show Next.js Server Component streaming hydration boundaries, WebSocket bidirectional frame ping-pong, and database mutation waterfalls.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Modern ESNext language features, Microtasks, Async/Await & Generator functions	Template <code>flow</code> : Animated state progression
Block 2	Days 6–10	Next.js 14 App Router: Server Components, Streaming Suspense & Route Handlers	Template <code>component-tree</code> : Animated state progression
Block 3	Days 11–15	Server Actions, Client component boundaries & Optimistic UI updates	Template <code>sequence</code> : Animated state progression
Block 4	Days 16–20	Type-safe ORM persistence: Prisma schemas, Relational joins & DB migrations	Template <code>table</code> : Animated state progression
Block 5	Days 21–25	Real-time communication: WebSocket duplex channels & Server-Sent Events (SSE)	Template <code>states</code> : Animated state progression
Block 6	Days 26–30	Production deployment: Edge middleware, Vercel serverless & Dockerized images	Template <code>compare</code> : Animated state progression

Capstone Milestone: Real-Time Collaborative Workspace: Full-stack application with live multi-user WebSocket editing, Prisma database synchronization, and optimistic UI transitions.

Course 14: IoT, Firmware & Embedded Systems

Catalog: course-iot-embedded

Difficulty: Intermediate | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: flow, states, cells, boxes, sequence, register-bits

Architectural Theme: Hardware Registers, Memory-Mapped I/O, Interrupts & FreeRTOS Schedulers

Pedagogical Whiteboard Model: Diagram microcontroller memory-mapped registers, bitwise mask operations, interrupt vector dispatching, and FreeRTOS task priority preemption.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	MCU architectures (ARM Cortex-M), Memory map & GPIO register addressing	Template flow : Animated state progression
Block 2	Days 6–10	Bitwise register manipulation: Bitmasking, Atomic Set/Reset (BSRR) & Pull-ups	Template states : Animated state progression
Block 3	Days 11–15	Timers, PWM signal generation & Hardware Interrupt Service Routines (ISR)	Template cells : Animated state progression
Block 4	Days 16–20	Analog-to-Digital Conversion (ADC): Quantization step sizing & Two-point scaling	Template boxes : Animated state progression
Block 5	Days 21–25	Serial protocols: UART baud rates, I2C 7-bit bus addressing & SPI clock polarity	Template sequence : Animated state progression
Block 6	Days 26–30	Real-Time Operating Systems: FreeRTOS preemptive scheduler, Queues & Semaphores	Template register-bits : Animated state progression

Capstone Milestone: Embedded Industrial Telemetry Controller: FreeRTOS firmware implementing multi-channel ADC sensor acquisition, lock-free ring buffers, and I2C peripheral telemetry.

Course 15: 3D Interactive Graphics & Avatar Animation

Catalog: course-3d-graphics

Difficulty: Advanced | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: tree-graph, boxes, flow, compare, bars

Architectural Theme: Vector Math, WebGL Render Pipeline, Shaders, Scene Graphs & Skeletal Rigging

Pedagogical Whiteboard Model: Visualize the 3D Model-View-Projection (MVP) matrix transformation pipeline, shader rasterization stages, and hierarchical skeletal bone transforms.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	3D Coordinate geometry: Cartesian vectors, Dot/Cross products & Matrix transforms	Template tree-graph : Animated state progression
Block 2	Days 6–10	WebGL rendering pipeline: Vertex Shaders, Fragment Shaders & Rasterization	Template boxes : Animated state progression
Block 3	Days 11–15	Three.js scene graph architecture: Object3D hierarchy, Meshes & Geometry buffers	Template flow : Animated state progression
Block 4	Days 16–20	Material physics & Lighting: Ambient, Point, Directional & Physically Based Rendering	Template compare : Animated state progression
Block 5	Days 21–25	Skeletal animation: Rigged bone hierarchies, Skinning weights & Keyframe tracks	Template bars : Animated state progression
Block 6	Days 26–30	Rendering optimizations: Draw call batching, Frustum culling & LOD geometries	Template tree-graph : Animated state progression

Capstone Milestone: Interactive 3D Avatar Animation Studio: WebGL-accelerated 3D viewport featuring custom PBR lighting shaders, skeletal bone transforms, and interactive orbital controls.

Course 16: Blockchain, Web3 & Smart Contracts

Catalog: course-blockchain-web3

Difficulty: Intermediate | **Duration:** 4 Weeks (30 Days / 6 Blocks) | **Quests:** 96 quests | **Allowed:** tree-graph, stack-queue, flow, sequence, compare, states

Architectural Theme: Cryptographic Hashing, Merkle Trees, EVM Gas Mechanics & Smart Contracts

Pedagogical Whiteboard Model: Show Merkle tree cryptographic proof verification, EVM memory/stack gas consumption charts, and reentrancy attack state transition barriers.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Cryptographic hash functions (SHA-256, Keccak), Asymmetric signatures & P2P networks	Template tree-graph : Animated state progression
Block 2	Days 6–10	Blockchain ledger structure: Block headers, Merkle Patricia Tries & Consensus	Template stack-queue : Animated state progression
Block 3	Days 11–15	Ethereum Virtual Machine (EVM): Execution stack, Storage slots & Gas accounting	Template flow : Animated state progression
Block 4	Days 16–20	Solidity contract development: Value types, Access modifiers & Event emissions	Template sequence : Animated state progression
Block 5	Days 21–25	Smart contract security: Checks-Effects-Interactions, Reentrancy & Integer safety	Template compare : Animated state progression
Block 6	Days 26–30	Decentralized frontend: Ethers.js / Viem, Metamask wallet RPC & Token transactions	Template states : Animated state progression

Capstone Milestone: Decentralized Escrow Protocol & dApp: Production Solidity smart contract with multi-signature release conditions, reentrancy guards, and reactive Web3 frontend.

Course 17: IoT Wireless Networks & Protocols

Catalog: `course-iot-network`

Difficulty: Intermediate | **Duration:** 4 Weeks (30 Days / 6 Blocks) | **Quests:** 96 quests | **Allowed:** `flow`, `sequence`, `states`, `table`, `compare`

Architectural Theme: Wireless RF Topologies, BLE GATT Profiles, LoRaWAN Chirps & MQTT Brokers

***Pedagogical Whiteboard Model:** Map BLE GATT Service/Characteristic hierarchies, LoRaWAN chirp spread spectrum uplinks, and MQTT publish/subscribe broker message routing.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Wireless RF propagation fundamentals, Frequencies (ISM bands) & Network topologies	Template <code>flow</code> : Animated state progression
Block 2	Days 6–10	Bluetooth Low Energy (BLE): GAP advertising, GATT Services & Characteristics	Template <code>sequence</code> : Animated state progression
Block 3	Days 11–15	IEEE 802.15.4 & Zigbee Mesh: Multi-hop routing & Self-healing network trees	Template <code>states</code> : Animated state progression
Block 4	Days 16–20	LoRa & LoRaWAN: Chirp Spread Spectrum (CSS), Gateway uplinks & Join procedures	Template <code>table</code> : Animated state progression
Block 5	Days 21–25	IoT Application protocols: MQTT QoS 0/1/2 handshakes, CoAP & Lightweight M2M	Template <code>compare</code> : Animated state progression
Block 6	Days 26–30	Ultra-low-power optimization: Deep sleep modes, Duty cycling & Energy budgets	Template <code>flow</code> : Animated state progression

Capstone Milestone: Low-Power LoRaWAN Environmental Gateway: Edge mesh network firmware routing environmental sensor telemetry through encrypted LoRaWAN packets to an MQTT cloud broker.

Course 18: Edge AI, DSP & TinyML Systems

Catalog: course-iot-edge-ai

Difficulty: Advanced | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: flow, bars, cells, table, compare

Architectural Theme: Digital Signal Processing, Feature Extraction, INT8 Quantization & TinyML

Pedagogical Whiteboard Model: Show sliding window sensor buffers, Fast Fourier Transform (FFT) spectrograms, and INT8 quantized weight matrices mapped to MCU memory arenas.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Edge intelligence paradigms, Microcontroller resource budgets (SRAM/Flash)	Template flow : Animated state progression
Block 2	Days 6–10	Digital Signal Processing (DSP): Discrete filtering, Windowing & Fast Fourier Transform	Template bars : Animated state progression
Block 3	Days 11–15	Time-series feature extraction: Spectral power, Zero-crossing rates & Peak detection	Template cells : Animated state progression
Block 4	Days 16–20	Model optimization for edge: Post-training INT8 quantization & Weight pruning	Template table : Animated state progression
Block 5	Days 21–25	TensorFlow Lite for Microcontrollers (TFLM): Memory arenas & Flatbuffer models	Template compare : Animated state progression
Block 6	Days 26–30	Real-time inference pipeline: Accelerometer gesture recognition & Anomaly detection	Template flow : Animated state progression

Capstone Milestone: Edge TinyML Gesture Recognition Engine: Microcontroller-deployed model classifying continuous IMU sensor streams in real-time under a 32 KB SRAM memory constraint.

Course 19: Industrial IoT Security & Device Lifecycle

Catalog: `course-iot-security`

Difficulty: Advanced | **Duration:** 4 Weeks (30 Days / 6 Blocks) | **Quests:** 96 quests | **Allowed:** `flow`, `sequence`, `states`, `compare`, `table`

Architectural Theme: Hardware Root of Trust, Secure Boot, Mutual TLS & Secure OTA Lifecycles

Pedagogical Whiteboard Model: *Diagram secure boot cryptographic signature verification chains, mTLS certificate exchanges, and dual-bank OTA firmware flash memory rollback safety.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Industrial IoT threat vectors, Purdue Enterprise Reference Model & Attack surfaces	Template <code>flow</code> : Animated state progression
Block 2	Days 6–10	Hardware Root of Trust: Secure Elements (ATECC608), TPMs & Cryptographic coprocessors	Template <code>sequence</code> : Animated state progression
Block 3	Days 11–15	Secure Boot architectures: Cryptographic hash validation & Anti-rollback eFuses	Template <code>states</code> : Animated state progression
Block 4	Days 16–20	Identity & Transport security: X.509 Device certificates & Mutual TLS (mTLS)	Template <code>compare</code> : Animated state progression
Block 5	Days 21–25	Secure Over-The-Air (OTA) updates: Dual-bank flash partitions & Rollback protection	Template <code>table</code> : Animated state progression
Block 6	Days 26–30	Industrial communication security: Encrypted Modbus-TCP, OPC UA & IEC 62443	Template <code>flow</code> : Animated state progression

Capstone Milestone: Hardened Industrial Edge Controller: Secure boot-verified firmware implementing mutual TLS cloud authentication and failsafe dual-bank OTA update recovery.

Course 20: Python Programming & Backend Systems

Catalog: course-python-backend

Difficulty: Intermediate | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: flow, sequence, boxes, table, states, compare

Architectural Theme: Python Asynchronous Systems, FastAPI, Pydantic & Distributed Task Queues

Pedagogical Whiteboard Model: Illustrate *asyncio* event loop coroutine yields, Pydantic JSON schema validations, and Celery asynchronous task distribution across Redis workers.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Advanced Python paradigms: Context managers, Decorators & Generator iterables	Template flow : Animated state progression
Block 2	Days 6–10	Asynchronous programming: asyncio event loops, Coroutine tasks & Non-blocking I/O	Template sequence : Animated state progression
Block 3	Days 11–15	FastAPI architecture: Path operations, Pydantic type models & Dependency Injection	Template boxes : Animated state progression
Block 4	Days 16–20	Database persistence: SQLAlchemy 2.0 Async Session, PostgreSQL & Alembic migrations	Template table : Animated state progression
Block 5	Days 21–25	Distributed task execution: Celery workers, Redis message brokers & Background tasks	Template states : Animated state progression
Block 6	Days 26–30	Production engineering: Docker containerization, Pytest unit tests & Rate limiters	Template compare : Animated state progression

Capstone Milestone: High-Performance Asynchronous FastAPI Service: Production microservice featuring async database connection pooling, background Celery processing, and OpenAPI contracts.

Course 21: Quantitative Engineering & Low-Latency Trading Systems Catalog: `course-quant-systems`

Difficulty: Advanced | **Duration:** 4 Weeks (30 Days / 6 Blocks) | **Quests:** 96 quests | **Allowed:** `stack-queue`, `table`, `bars`, `flow`, `cells`, `compare`

Architectural Theme: Limit Order Books, VWAP Slicing, Low-Latency Sockets & Microsecond Systems

***Pedagogical Whiteboard Model:** Show Limit Order Book (LOB) bid/ask price queues, execution algorithms slicing volume over time, and kernel-bypass zero-copy socket buffer flows.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Market microstructure: Limit Order Books (LOB), Depth of market & Bid-Ask spreads	Template <code>stack-queue</code> : Animated state progression
Block 2	Days 6–10	Financial timeseries analytics: Returns, Rolling volatility & Log price transforms	Template <code>table</code> : Animated state progression
Block 3	Days 11–15	Algorithmic execution strategies: VWAP, TWAP, Implementation shortfall & Slippage	Template <code>bars</code> : Animated state progression
Block 4	Days 16–20	Low-latency system engineering: Zero-copy buffers, Cache locality & Kernel bypass	Template <code>flow</code> : Animated state progression
Block 5	Days 21–25	Risk modeling & Portfolio metrics: Value at Risk (VaR), Sharpe Ratio & Max Drawdown	Template <code>cells</code> : Animated state progression
Block 6	Days 26–30	High-frequency backtesting engine: Event-driven matching simulation & FIX protocol	Template <code>compare</code> : Animated state progression

Capstone Milestone: Microsecond Limit Order Book Matching Engine: High-frequency order matching engine simulating sub-millisecond price-time priority execution and real-time PnL tracking.

Course 22: Digital Accounting & Taxation (B.Com / BBA)

Catalog: course-digital-accounting

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: table, flow, compare, bars, ledger-sheet

Architectural Theme: Double-Entry Invariants, Ledger Balancing, GST Taxation & ERP Compliance

Pedagogical Whiteboard Model: Draw double-entry debit/credit ledger columns maintaining balancing equality, 3-way invoice matching flows, and GST tax credit cascading waterfalls.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Accounting equation fundamentals: Assets = Liabilities + Equity & Business Entity rule	Template table : Animated state progression
Block 2	Days 6–10	Golden rules of accounting: Real, Personal & Nominal accounts & Journal posting	Template flow : Animated state progression
Block 3	Days 11–15	Ledger posting: T-Accounts balancing, Closing balances & Trial Balance checksums	Template compare : Animated state progression
Block 4	Days 16–20	Financial reporting: Trading Account, Profit & Loss statement & Balance Sheet layout	Template bars : Animated state progression
Block 5	Days 21–25	Taxation architecture: GST Input Tax Credit cascading & TDS statutory deductions	Template ledger-sheet : Animated state progression
Block 6	Days 26–30	Modern ERP workflows: Automated bank reconciliations & Suspense account resolution	Template table : Animated state progression

Capstone Milestone: Automated Corporate Accounting & GST Audit Engine: End-to-end digital accounting ledger with automated journal entries, balancing trial balances, and GST compliance validation.

Course 23: Business Finance & Investment Management (B.Com / BBA)

Catalog: course-finance-investment

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: table, bars, flow, compare

Architectural Theme: Time Value of Money, DCF Valuation, WACC Capital Structures & Portfolio Risk

Pedagogical Whiteboard Model: Visualize Discounted Cash Flow (DCF) timelines, DuPont 3-stage ROE breakdowns, and debt vs equity cost weights in WACC capital structure models.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Time Value of Money (TVM): Present Value, Future Value & Compounding schedules	Template table : Animated state progression
Block 2	Days 6–10	Capital budgeting evaluation: Net Present Value (NPV), IRR & Payback period hurdles	Template bars : Animated state progression
Block 3	Days 11–15	Cost of Capital: Weighted Average Cost of Capital (WACC) & Optimal Debt-Equity mix	Template flow : Animated state progression
Block 4	Days 16–20	Working capital dynamics: Cash Conversion Cycle, Operating cycle & Liquidity ratios	Template compare : Animated state progression
Block 5	Days 21–25	Corporate Valuation: DCF terminal values, Multiples analysis & DuPont 3-stage ROE	Template table : Animated state progression
Block 6	Days 26–30	Modern Portfolio Theory: Efficient frontier, Sharpe ratio & Beta risk sensitivity	Template bars : Animated state progression

Capstone Milestone: Corporate Financial Valuation & Investment Model: Comprehensive DCF valuation model with dynamic WACC sensitivity matrices, scenario planning, and working capital forecasts.

Course 24: Business Analytics & Decision Intelligence (B.Com / BBA / MBA)

Catalog: course-business-analytics

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: bars, table, flow, compare

Architectural Theme: Statistical Distributions, Pareto ABC Analysis, Regression & Executive KPIs

Pedagogical Whiteboard Model: Show descriptive distribution curves, Pareto 80/20 ABC classification breakdowns, and linear regression trendlines against executive scorecard KPIs.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Data-driven decision framework: Descriptive statistics, Mean, Median & Standard deviation	Template bars : Animated state progression
Block 2	Days 6–10	Data distributions: Normal curves, Skewness, Outlier detection & Interquartile range	Template table : Animated state progression
Block 3	Days 11–15	Business KPI architecture: Balanced scorecards, Attribution & Performance variance	Template flow : Animated state progression
Block 4	Days 16–20	Predictive analytics: Simple & Multiple OLS Linear regression & Trend forecasting	Template compare : Animated state progression
Block 5	Days 21–25	Classification & Segmentation: Customer RFM analysis & Logistic regression odds	Template bars : Animated state progression
Block 6	Days 26–30	Executive data storytelling: Automated KPI dashboards & Decision intelligence reports	Template table : Animated state progression

Capstone Milestone: Executive Business Intelligence & Decision Dashboard: End-to-end analytics report synthesizing customer segmentation, revenue regression forecasts, and automated KPI alert metrics.

Course 25: Marketing & Brand Management (B.Com / BBA / MBA)

Catalog: course-marketing-branding

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: flow, compare, table, bars

Architectural Theme: Strategic Frameworks, Consumer Psychology, Brand Equity & PLC Lifecycles

Pedagogical Whiteboard Model: Map customer journey stages, Keller brand equity pyramid levels, BCG growth-share matrix quadrants, and omnichannel campaign ROI bar charts.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Strategic marketing foundations: 4Ps Marketing Mix, STP & Porter’s Five Forces	Template flow : Animated state progression
Block 2	Days 6–10	Consumer psychology: Cognitive decision journey, Perception biases & Social proof	Template compare : Animated state progression
Block 3	Days 11–15	Brand identity architecture: Brand personality, Archetypes & Keller’s Brand Equity	Template table : Animated state progression
Block 4	Days 16–20	Product portfolio management: Boston Consulting Group (BCG) Matrix & Product Life Cycle	Template bars : Animated state progression
Block 5	Days 21–25	Integrated Marketing Communications (IMC): Omnichannel messaging & Content strategies	Template flow : Animated state progression
Block 6	Days 26–30	Marketing performance measurement: Customer Lifetime Value (CLV) & Brand sentiment	Template compare : Animated state progression

Capstone Milestone: Comprehensive Brand Strategy & Go-To-Market Blueprint: Multi-channel brand positioning framework complete with customer persona journeys, messaging matrices, and ROI forecasts.

Course 26: Digital Marketing & Growth Strategy (B.Com / BBA / MBA)

Catalog: course-digital-marketing

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: flow, bars, table, compare, funnel

Architectural Theme: Acquisition Funnels, Ad Auction Rankings, SEO Architectures & CAC/LTV Unit Economics

Pedagogical Whiteboard Model: Diagram conversion funnel drop-off waterfalls, ad auction rank bid vs quality score calculations, and CAC payback period bar comparisons.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Digital acquisition landscape: Inbound vs Outbound & Customer acquisition funnels	Template flow : Animated state progression
Block 2	Days 6–10	Search Engine Optimization (SEO): Crawl budgets, Technical SEO, On-page & Backlinks	Template bars : Animated state progression
Block 3	Days 11–15	Search Engine Marketing (SEM): Google Ads auction mechanics, Quality Score & CPC bidding	Template table : Animated state progression
Block 4	Days 16–20	Paid Social Advertising: Meta Ad pixel tracking, Lookalike audiences & ROAS targets	Template compare : Animated state progression
Block 5	Days 21–25	Email marketing automation: Behavioral drip sequences, Open rate & Churn telemetry	Template funnel : Animated state progression
Block 6	Days 26–30	Growth analytics: Multi-touch attribution models, A/B testing & CAC-to-LTV payback	Template flow : Animated state progression

Capstone Milestone: Growth Marketing & Performance Campaign Architecture: Complete multi-channel digital acquisition campaign with SEO keyword architecture, ad auction bidding models, and CAC/LTV funnels.

Course 27: E-Commerce & Digital Business (B.Com / BBA / MBA)

Catalog: `course-ecommerce-digital-biz`

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: `flow`, `table`, `sequence`, `bars`, `compare`, `funnel`

Architectural Theme: SKU Variant Matrices, Volumetric Logistics, Checkout Gateways & Marketplace Scaling

Pedagogical Whiteboard Model: Illustrate e-commerce checkout conversion funnels, payment gateway authorization handshakes, and dimensional weight shipping cost calculations.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	E-Commerce business models: Direct-to-Consumer (D2C), B2B & Multi-vendor marketplaces	Template <code>flow</code> : Animated state progression
Block 2	Days 6–10	Product catalog engineering: Parent-child SKU variants, Attributes & Inventory sync	Template <code>table</code> : Animated state progression
Block 3	Days 11–15	Checkout optimization: Cart abandonment mitigation, Payment gateways & Webhooks	Template <code>sequence</code> : Animated state progression
Block 4	Days 16–20	Supply chain fulfillment: 3PL logistics, Dimensional weight & Last-mile delivery	Template <code>bars</code> : Animated state progression
Block 5	Days 21–25	Marketplace scaling: Amazon Buy Box algorithms, Seller metrics & Commission fees	Template <code>compare</code> : Animated state progression
Block 6	Days 26–30	Cross-border digital commerce: Currency localization, Duties, Taxes & Compliance	Template <code>funnel</code> : Animated state progression

Capstone Milestone: Scalable E-Commerce Operations Blueprint: Full operational framework for high-volume digital commerce, including SKU catalog architecture, checkout conversion funnels, and logistics routing.

Course 28: Entrepreneurship & Business Management (B.Com / BBA / MBA)

Catalog: course-entrepreneurship-biz-mgmt

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: flow, table, compare, bars

Architectural Theme: Business Model Canvas, TAM/SAM/SOM Sizing, Cap Table Dilution & Runway Dynamics

Pedagogical Whiteboard Model: Show Business Model Canvas component linkages, bottom-up TAM/SAM/SOM concentric circles, equity dilution round-by-round cap tables, and monthly burn rate runway curves.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Entrepreneurial mindset, Problem-Solution fit & Lean Startup iterative hypothesis	Template flow : Animated state progression
Block 2	Days 6–10	Business Model Canvas (BMC): Revenue streams, Cost structures & Value propositions	Template table : Animated state progression
Block 3	Days 11–15	Market opportunity sizing: Total Addressable Market (TAM, SAM, SOM) calculation	Template compare : Animated state progression
Block 4	Days 16–20	Startup financial modeling: Unit economics, Burn rate & Cash runway forecasting	Template bars : Animated state progression
Block 5	Days 21–25	Venture financing mechanics: Term sheets, Convertible notes, SAFEs & Cap table dilution	Template flow : Animated state progression
Block 6	Days 26–30	Operational scaling: Hiring frameworks, Agile organizational design & Milestone gates	Template table : Animated state progression

Capstone Milestone: Venture Creation & Investor-Ready Business Plan: Comprehensive startup plan featuring validated Business Model Canvas, bottom-up market sizing, unit economics, and 5-year pro forma financials.

Course 29: Sales, Customer Success & CRM (B.Com / BBA / MBA) Catalog: `course-sales-crm-success`
Difficulty: Beginner | **Duration:** 4 Weeks (30 Days / 6 Blocks) | **Quests:** 96 quests | **Allowed:** `flow`, `table`, `states`, `bars`, `compare`, `funnel`

Architectural Theme: BANT/MEDDIC Qualification, Sales Velocity, CRM Workflows & Customer Retention

Pedagogical Whiteboard Model: *Diagram sales velocity formula components, MEDDIC qualification scorecards, lead-to-opportunity state progression, and customer health score distributions.*

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Modern B2B sales methodologies: Consultative selling, Challenger sales & Inbound leads	Template <code>flow</code> : Animated state progression
Block 2	Days 6–10	Opportunity qualification frameworks: BANT (Budget, Authority, Need, Timing) & MEDDIC	Template <code>table</code> : Animated state progression
Block 3	Days 11–15	Sales pipeline mechanics: Deal stages, Probability weightings & Sales Velocity formula	Template <code>states</code> : Animated state progression
Block 4	Days 16–20	CRM systems architecture: Lead routing workflows, Activity tracking & Pipeline automation	Template <code>bars</code> : Animated state progression
Block 5	Days 21–25	Commercial negotiation: Objection handling frameworks, Value selling & Closing tactics	Template <code>compare</code> : Animated state progression
Block 6	Days 26–30	Customer Success & Retention: Net Retention Rate (NRR), Churn telemetry & Health scores	Template <code>funnel</code> : Animated state progression

Capstone Milestone: Enterprise B2B Sales & CRM Operational Playbook: Configured CRM pipeline architecture with automated MEDDIC qualification scoring, sales velocity tracking, and retention playbooks.

Course 30: Operations, Supply Chain & Business Compliance

(B.Com / BBA / MBA)

Catalog: course-operations-supplychain-compliance

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: flow, table, bars, compare

Architectural Theme: SIPOC Value Stream Mapping, EOQ Inventory Models, Kraljic Matrices & Quality Control

Pedagogical Whiteboard Model: Draw SIPOC value stream process flows, Economic Order Quantity (EOQ) cost trade-off curves, Kraljic matrix supplier quadrants, and Six Sigma defect distributions.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Operations strategy: Process architecture, Capacity planning & Bottleneck identification	Template flow : Animated state progression
Block 2	Days 6–10	Value Stream Mapping: SIPOC diagrams, Lead time reduction & Waste elimination (Muda)	Template table : Animated state progression
Block 3	Days 11–15	Inventory optimization: Economic Order Quantity (EOQ), Safety stock & Reorder points	Template bars : Animated state progression
Block 4	Days 16–20	Strategic procurement: Kraljic purchasing portfolio matrix & Vendor SLA evaluations	Template compare : Animated state progression
Block 5	Days 21–25	Total Quality Management (TQM): Six Sigma DMAIC cycle, Statistical process control & Kaizen	Template flow : Animated state progression
Block 6	Days 26–30	Business compliance: Supply chain ethics, ISO certifications & Regulatory audit governance	Template table : Animated state progression

Capstone Milestone: End-to-End Supply Chain Optimization Plan: Operations master plan featuring value stream mapping, automated EOQ inventory replenishment models, and supplier quality audits.

Course 31: AI & Digital Transformation for Business (B.Com / BBA / MBA)

Catalog: course-ai-digital-transformation

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: workflow, flow, table, compare, bars

Architectural Theme: Enterprise AI Value Equations, Robotic Process Automation & AI Governance

Pedagogical Whiteboard Model: Illustrate enterprise AI value creation frameworks, RPA bot automation process flows, AI governance risk-level classifications, and digital maturity phase roadmaps.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Digital Transformation imperatives: Legacy modernization vs Digital-first paradigms	Template workflow : Animated state progression
Block 2	Days 6–10	Enterprise AI Value Equation: Cost reduction, Revenue acceleration & Customer experience	Template flow : Animated state progression
Block 3	Days 11–15	Robotic Process Automation (RPA): Workflow orchestration & Repetitive task bot automation	Template table : Animated state progression
Block 4	Days 16–20	Generative AI workplace deployment: Departmental copilot adoption & Prompt standards	Template compare : Animated state progression
Block 5	Days 21–25	AI Governance & Ethics: Bias auditing, Privacy protection, IP safety & EU AI Act compliance	Template bars : Animated state progression
Block 6	Days 26–30	Organizational Change Management: Cultural readiness, Upskilling roadmaps & ROI dashboards	Template workflow : Animated state progression

Capstone Milestone: Enterprise AI Transformation Roadmap & Governance Charter: Multi-year digital transformation strategic roadmap with departmental RPA automation blueprints and ethical AI policies.

Course 32: Computer Literacy, Digital Productivity & OS Fundamentals

Catalog: course-computer-fundamentals

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: flow, boxes, tree-graph, table, compare

Architectural Theme: Von Neumann Architecture, POSIX File Permissions, CLI Pipelines & TCP/IP Networking

Pedagogical Whiteboard Model: Diagram Input-Process-Output CPU bus flows, Unix command pipe data streams, file system inode permission trees, and TCP/IP 4-layer packet traversals.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Computer architecture: CPU registers, Arithmetic Logic Unit (ALU), RAM & Bus topology	Template flow : Animated state progression
Block 2	Days 6–10	Operating Systems fundamentals: Kernel mode vs User mode, System calls & Process threads	Template boxes : Animated state progression
Block 3	Days 11–15	POSIX Command Line: Terminal navigation, File permissions (chmod/chown) & Pipe redirection	Template tree-graph : Animated state progression
Block 4	Days 16–20	Memory and storage hierarchy: L1/L2/L3 CPU Caches, Virtual memory paging & RAID arrays	Template table : Animated state progression
Block 5	Days 21–25	Computer networking basics: TCP/IP 4-layer model, IPv4 subnetting, DNS & Port routing	Template compare : Animated state progression
Block 6	Days 26–30	Digital security & System health: Process monitoring, SSH keys & Preventive maintenance	Template flow : Animated state progression

Capstone Milestone: Systems Administration & Networking Verification Suite: Practical command-line and systems automation project demonstrating process monitoring, shell scripting, and network diagnostics.

Course 33: Everyday AI Literacy & Prompt Engineering

Catalog: course-ai-prompt-literacy

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: flow, compare, boxes, table, bars

Architectural Theme: Prompt Engineering Frameworks, Chain-of-Thought Reasoning & Fact Verification

Pedagogical Whiteboard Model: Show prompt before/after optimization comparisons, Chain-of-Thought reasoning step paths, temperature sampling spaces, and fact-checking verification scorecards.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Generative AI mechanics: Probabilistic language modeling, Tokens & Prediction spaces	Template flow : Animated state progression
Block 2	Days 6–10	Prompt structuring foundations: Context, Persona, Instruction & Constraint frameworks	Template compare : Animated state progression
Block 3	Days 11–15	Advanced prompting techniques: Few-shot examples, Chain-of-Thought (CoT) & Decomposition	Template boxes : Animated state progression
Block 4	Days 16–20	Multimodal AI capabilities: Text-to-image prompting, Audio transcription & Visual analysis	Template table : Animated state progression
Block 5	Days 21–25	Workplace productivity automation: Research synthesis, Executive drafting & Data extraction	Template bars : Animated state progression
Block 6	Days 26–30	Cognitive guardrails: Hallucination identification, Source grounding & Ethical evaluation	Template flow : Animated state progression

Capstone Milestone: Master Prompt Engineering Playbook: Production-grade collection of verified, structured prompt templates for workplace research, communication, data analysis, and validation.

Course 34: Excel & Data Analysis Fundamentals

Catalog: course-excel-data-viz

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: table, cells, bars, compare, flow

Architectural Theme: 2D Coordinate Grids, Logical Evaluation, XLOOKUP Intersections & Executive Pivot Dashboards

Pedagogical Whiteboard Model: Illustrate 2D grid cell referencing, XLOOKUP array search traversals, Pivot Table multidimensional aggregations, and executive waterfall chart components.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Spreadsheet grid architecture: 2D coordinate referencing, Absolute (\$) vs Relative locks	Template table : Animated state progression
Block 2	Days 6–10	Statistical functions: SUM, AVERAGE, COUNT, MIN/MAX & Weighted average calculations	Template cells : Animated state progression
Block 3	Days 11–15	Logical evaluations: Single IF, Nested IF statements & Multi-condition AND/OR formulas	Template bars : Animated state progression
Block 4	Days 16–20	Text sanitization & Date calculations: TRIM, CLEAN, TEXTJOIN, CONCAT & DATEDIF math	Template compare : Animated state progression
Block 5	Days 21–25	Lookup & Reference functions: Classic VLOOKUP vs Modern XLOOKUP & 2-Way INDEX-MATCH	Template flow : Animated state progression
Block 6	Days 26–30	Data visualization & Reporting: Pivot Tables, Multi-dimensional slicing & Executive charts	Template table : Animated state progression

Capstone Milestone: Executive Financial & Operational Excel Model: Fully automated corporate spreadsheet workbook featuring dynamic XLOOKUP models, Pivot Table aggregations, and executive visual dashboards.

Course 35: Git, GitHub & Version Control Basics

Catalog: course-git-version-control

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: tree-graph, flow, states, compare, boxes

Architectural Theme: Directed Acyclic Graphs, 3-Tier Git Areas, 3-Way Merges & Collaborative PRs

Pedagogical Whiteboard Model: Draw 3-tier Git architecture (Working directory -> Staging -> Repository), Directed Acyclic Graph (DAG) commit trees, and 3-way merge conflict marker resolutions.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Version control architecture: Working Directory, Staging Area (Index) & Local Repository	Template tree-graph : Animated state progression
Block 2	Days 6–10	Core versioning operations: Commit SHA hashes, Atomic commits, Git log & History inspection	Template flow : Animated state progression
Block 3	Days 11–15	Branching dynamics: Pointer movements, Fast-forward merges & Feature branch isolation	Template states : Animated state progression
Block 4	Days 16–20	Branch reconciliation: 3-way merge algorithms, Merge conflict resolution & Interactive rebase	Template compare : Animated state progression
Block 5	Days 21–25	Remote collaboration: Git remotes, Push/Pull mechanics, Upstream tracking & Fetching	Template boxes : Animated state progression
Block 6	Days 26–30	GitHub team workflows: Pull requests, Code review etiquettes, Branch protection & Tags	Template tree-graph : Animated state progression

Capstone Milestone: Multi-Branch Collaborative Repository Simulation: Realistic engineering project demonstrating trunk-based branching, clean rebase histories, resolved merge conflicts, and peer-reviewed PRs.

Course 36: Professional Tech Communication & Interview Mastery

Catalog: course-softskills-communication

Difficulty: Beginner | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: flow, compare, table, states

Architectural Theme: Minto Pyramid Structured Messaging, Active Listening, STAR Method & Tech Interviews

Pedagogical Whiteboard Model: Diagram Minto Pyramid top-down communication hierarchies, STAR method (Situation, Task, Action, Result) storytelling flows, and email tone before/after comparison tables.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Foundations of technical communication: Clarity, Conciseness & The BLUF principle	Template flow : Animated state progression
Block 2	Days 6–10	Structured thinking models: Minto Pyramid, MECE principle & Executive summaries	Template compare : Animated state progression
Block 3	Days 11–15	Professional written correspondence: High-impact emails, Slack etiquette & Documentation	Template table : Animated state progression
Block 4	Days 16–20	Verbal presentations: Agile standup updates, Technical demos & Slide design principles	Template states : Animated state progression
Block 5	Days 21–25	Interpersonal dynamics: Active listening, Constructive feedback loops & Conflict de-escalation	Template flow : Animated state progression
Block 6	Days 26–30	Interview mastery: The STAR method, Behavioral question responses & Salary negotiation	Template compare : Animated state progression

Capstone Milestone: Executive Career Portfolio & Interview Masterclass: Comprehensive career communication asset package featuring structured technical project demos, STAR behavioral interview decks, and executive writing samples.

Course 37: Natural Language Processing & Computational Linguistics

Catalog: course-nlp

Difficulty: Advanced | Duration: 4 Weeks (30 Days / 6 Blocks) | Quests: 96 quests | Allowed: letters, cells, table, flow, bars, tree-graph

Architectural Theme: Tokenization, TF-IDF Vectors, Word2Vec Semantics & Transformer Self-Attention

Pedagogical Whiteboard Model: Show token sequence sliding windows, word embedding vector spaces with cosine distance, and Transformer Query-Key-Value self-attention matrix products.

Block	Days	Curriculum Topic & Real Lessons	Visual Engine Architecture
Block 1	Days 1–5	Text preprocessing pipelines: Unicode normalization, Tokenization, Stemming & Lemmatization	Template letters : Animated state progression
Block 2	Days 6–10	Statistical representation: Bag-of-Words, N-grams & Term Frequency-Inverse Document Frequency	Template cells : Animated state progression
Block 3	Days 11–15	Distributed vector embeddings: Continuous Bag of Words (CBOW), Skip-gram & Word2Vec geometry	Template table : Animated state progression
Block 4	Days 16–20	Sequence modeling: Recurrent Neural Networks (RNN), Hidden states & LSTM/GRU gating mechanisms	Template flow : Animated state progression
Block 5	Days 21–25	Transformer architecture: Scaled dot-product Self-Attention, Multi-head matrices & Positional encodings	Template bars : Animated state progression
Block 6	Days 26–30	Computational language applications: Named Entity Recognition (NER), Sentiment scoring & Summarization	Template tree-graph : Animated state progression

Capstone Milestone: Production Natural Language Processing Engine: End-to-end computational linguistic pipeline performing text preprocessing, TF-IDF/Dense embedding extraction, and sentiment/NER classification.

6. Execution & Phased Implementation Roadmap

The generation and verification of visual assets across all 37 individual 1-month courses follows an atomic, verifiable protocol:

Phase	Scope & Objectives	Verification Gate & Deliverables
Phase 0: Safety & Scope	Protected file locks, Gitleaks secret verification, course directory initialization across all 37 prefixes.	Pre-flight checks pass; zero uncommitted modifications; clean working directory.
Phase 1: Engine Buildout	Register 3 extended templates (<code>register-bits</code> , <code>ledger-sheet</code> , <code>funnel</code>); wire financial ledger and bitwise binding adapters.	All 17 templates compile in <code>VisualStage.tsx</code> ; TypeScript typechecks clean.
Phase 2: 5-Day Sprints	Generate 222 atomic 5-day blocks across the 37 courses using the 5-Step Loop.	Every block outputs <code>PASS 5/6</code> or <code>PASS 6/6</code> on Gate v2 before atomic git commit.
Phase 3: Student Simulation	Full-matrix runtime audit across all 1,110 lesson days and 6,660 parts using <code>simulate-student-view.mts</code> .	100% matrix pass rate (0 fatal errors, 0 broken bindings, 0 layout crashes).
Phase 4: Owner Sign-Off	Sample inspection of 3 lessons per cluster on mobile (390px) and desktop (1440px) viewports in light and dark modes.	Official release switches flipped in <code>enabledCourses.ts</code> .

PinIT Career OS • Publication-Grade Visuals Master Architecture • 37 Courses • 100% Gate v2 Compliant